



Department of Electronics Engineering

ELEC334 – Microprocessors
Homework 1 Report

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1. Introduction

This report presents the design and implementation of a line-following robot developed as part of the ELEC334 Microprocessors course. The robot system is built around the STM32F103 microcontroller and integrates optical sensors, an L293D motor driver, and a buzzer module. The purpose of this project is to demonstrate embedded system design principles through sensor-based motion control, battery monitoring, and PWM-based motor actuation.

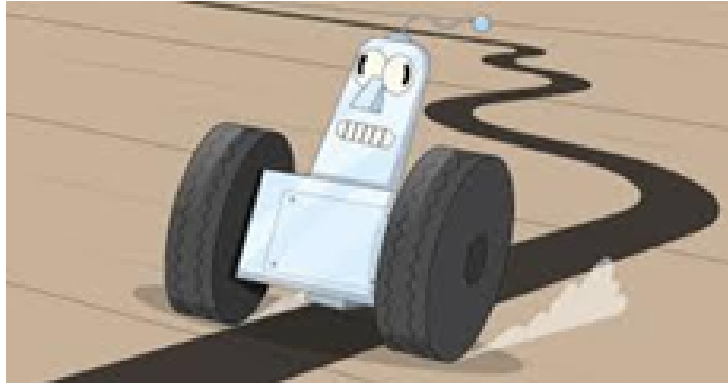


Figure 1. Robot

STM32F103C8T6 Board Components and Peripherals: In this section, the STM32F103C8T6 (Blue Pill) development board is examined in detail. All the integrated circuits (ICs), onboard peripherals, and their electrical connections with the microcontroller are identified and explained

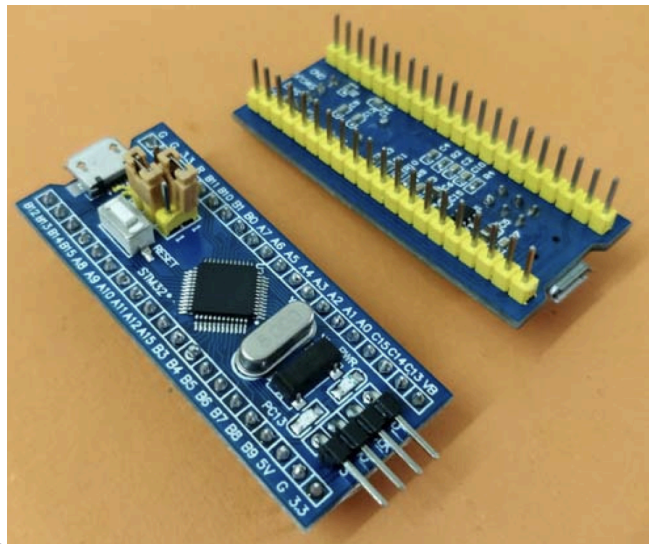


Figure 2. STM32F103C8T6

STM32F103C8T6: This section focuses on introducing the STM32F103C8T6 microcontroller and outlining its essential properties. Built around the ARM Cortex-M3 32-bit core with a RISC architecture, it operates at clock speeds of up to 72 MHz, providing a balance between performance and energy efficiency. The chip is equipped with 64 KB of Flash memory for program storage and 20 KB of SRAM for data handling, all powered by a 3.3 V supply. In terms of functionality, the STM32F103C8T6 offers a versatile set of peripherals. It supports multiple communication interfaces—including three USART ports for serial transmission, two I²C buses for device interconnection, three SPI channels for rapid data transfer, and a CAN interface designed for robust industrial communication. Furthermore, it incorporates a 12-bit Analog-to-Digital Converter with 10 input channels, timers for PWM generation and signal capture, and GPIO ports A, B, and C that enable direct interaction with sensors, switches, or LEDs. The device also provides USB 2.0 full-speed connectivity, making it suitable for embedded applications requiring efficient communication and control.

2. System Overview

The robot operates based on input signals obtained from two infrared optical sensors labeled as the left and right sensors. Each sensor produces a logic high (1) when it detects the black line and a logic low (0) when it detects the white surface. The system determines motion direction according to these sensor readings.

2.1 Line Following Algorithm

The control algorithm defines four primary behaviors based on sensor input:

- Left Turn: Left sensor detects black, right sensor detects white.
- Right Turn: Right sensor detects black, left sensor detects white.
- Forward Motion: Both sensors detect white.
- Stop: Both sensors detect black.

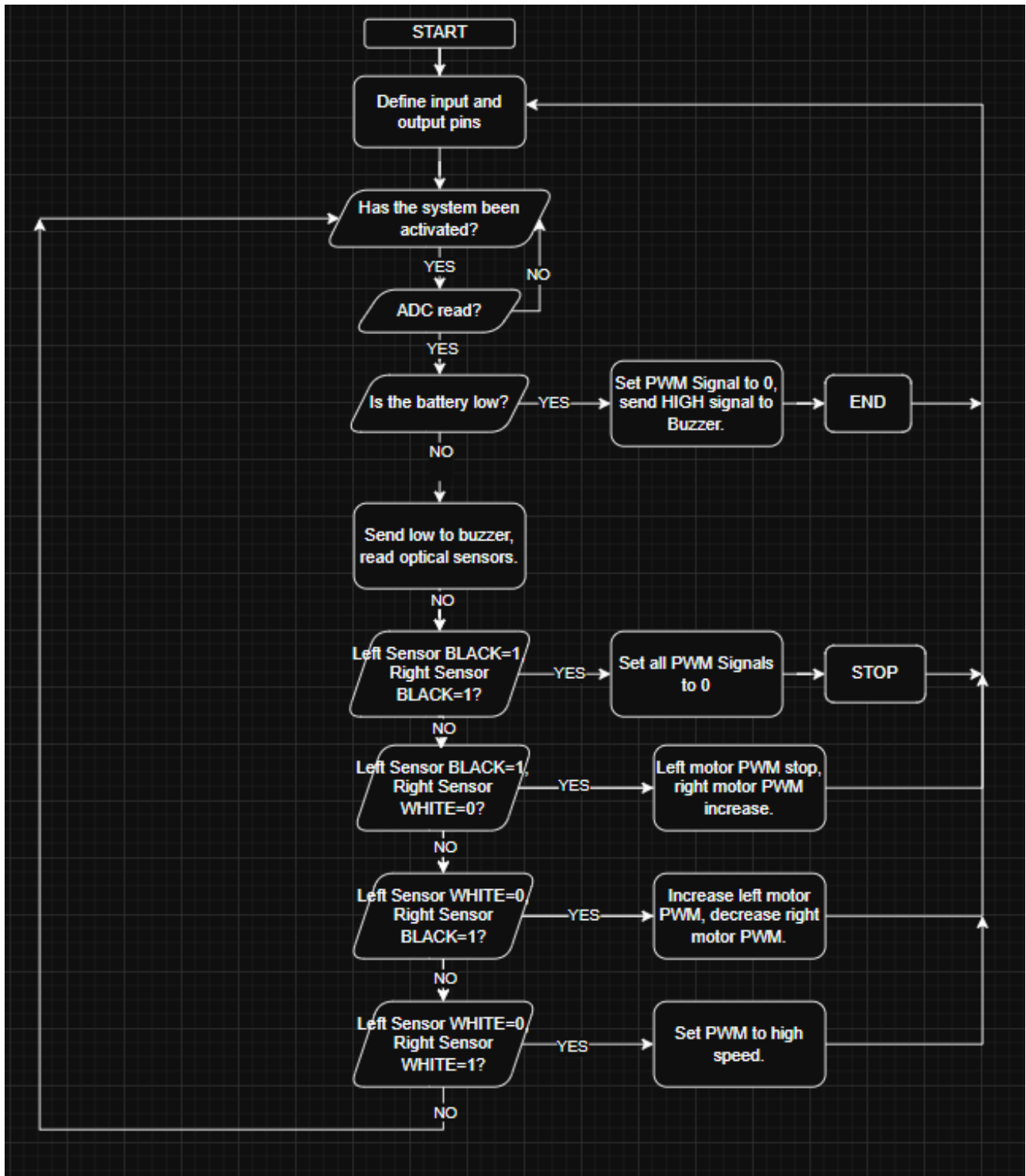


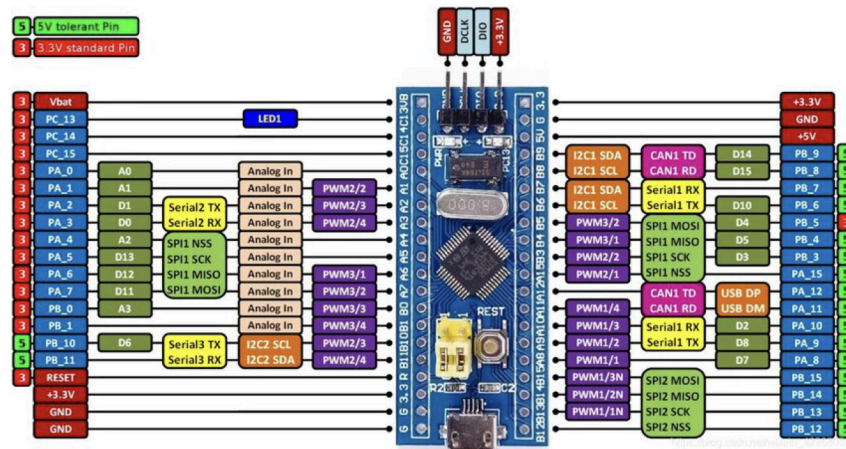
Figure 3. Flowchart

2.2 Speed and Power Management

Motor speed is regulated using Pulse Width Modulation (PWM) signals generated by the STM32 timer peripherals. Additionally, the system dynamically adjusts motor speed according to battery voltage, ensuring consistent performance as the battery discharges.

3. Hardware Design

The hardware design incorporates the STM32F103 'Blue Pill' microcontroller as the main control unit. Peripheral components include optical sensors for line detection, an L293D motor driver for dual DC motor control, and a buzzer for low-battery alerts.



Peripheral	STM32 Pin(s)	Function
User LED	PC13	Output (active low)
Reset Button	NRST	Manual reset
BOOT0 Jumper	BOOT0	Boot configuration
USB-UART (CH340G)	PA9 (TX1), PA10 (RX1)	Serial communication with PC
Voltage Regulator (AMS1117)	Input: 5 V, Output: 3.3 V	Power supply for MCU
8 MHz Crystal	OSC_IN (PA0) / OSC_OUT (PA1)	System clock
32.768 kHz Crystal	PC14, PC15	RTC clock
Power Input	5 V (USB or pin), GND	Board power source
3.3 V Output	VCC pin	For powering sensors/modules

Figure 4. STM32 PINS

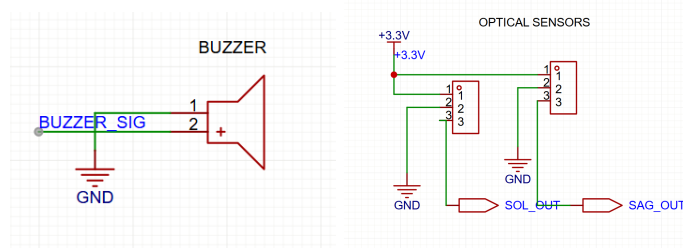


Figure 5. Buzzer with Sensors

3.1 Pin Configuration

The pin assignments for the system are listed as follows:

Power Input: 3.3V regulated (VDD), Ground (VSS),VBAT

Sensors (Digital Inputs):

- PA0: Left Optical Sensor
- PA1: Right Optical Sensor

Motor Driver (PWM and Directional Control):

- PA8 (TIM1_CH1): Left Motor Speed (PWM_L)
- PA9 (TIM1_CH2): Right Motor Speed (PWM_R)
- PB0, PB1: Left Motor Direction (IN_L1, IN_L2)
- PB2, PB4: Right Motor Direction (IN_R3, IN_R4)

Battery Voltage Monitoring:

- PA4 (ADC1_IN4): Battery Voltage Divider Output

Buzzer Output:

- PA5: Buzzer Signal Output

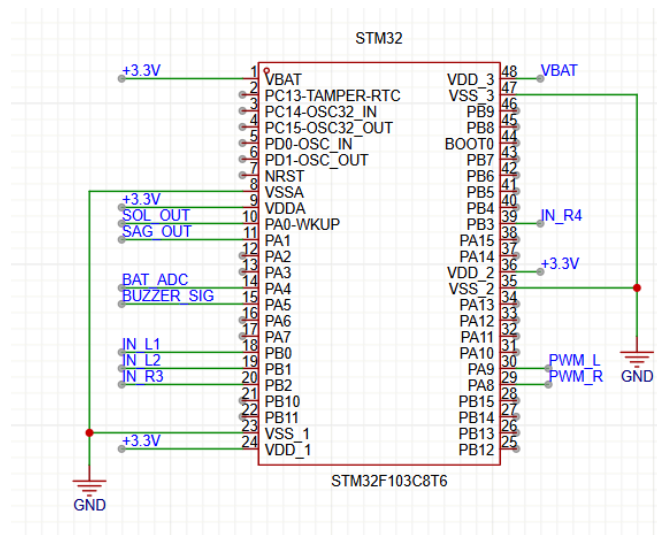


Figure 6. STM32

3.2 Electrical Characteristics

The system uses a single-cell Li-Ion battery (3.4V–4.2V) as its main power source. An AMS1117-3.3 LDO regulator converts this to 3.3V for logic-level circuits, including the STM32, L293D logic supply (VSS), and optical sensors. The motor supply (VS) is connected directly to the unregulated battery line to maximize output power.

Voltage thresholds are defined as follows:

- Fully Charged: 4.2V
- Low Battery Warning: 3.4V

4. Battery Voltage Detection via ADC

Since the STM32 ADC input cannot exceed 3.3V while the battery voltage may reach 4.2V, a voltage divider network is employed. Two 10kΩ resistors are connected in series between the battery terminal (VBAT) and ground, and the midpoint voltage is fed into PA4 (ADC1_IN4). This configuration halves the measured voltage, protecting the microcontroller input. In software, the ADC reading is multiplied by two to reconstruct the actual battery voltage. If the measured voltage drops below 3.4V, the system halts motor operation and activates the buzzer on PA5 to indicate a low-battery condition.

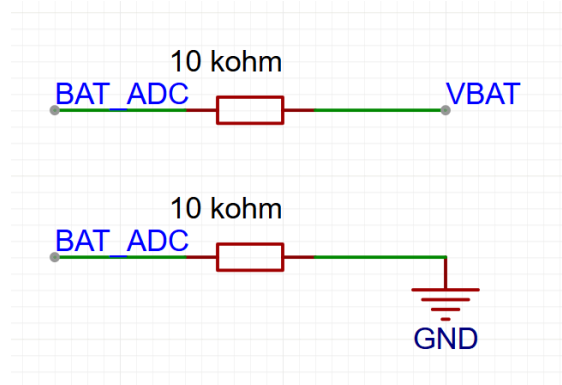


Figure 7. Battery Connection

4.1 Power Management and Regulator

The project's power management is supplied by a Li-Ion battery, which provides a variable voltage ranging from 3.4V to 4.2V. However, sensitive digital components, such as the system's main controller (STM32F103 microcontroller) and the optical sensors, require a stable 3.3V supply.

To meet this requirement, an **AMS1117-3.3** Low-Dropout Regulator (LDO) is utilized in the schematic design. This regulator takes the variable VBAT voltage from the battery and converts it into a stable 3.3V output (the +3.3V rail). This 3.3V rail is used to power all VDD

pins of the STM32F103, the sensors, and the logic (VSS) pin of the L293D motor driver. This ensures the stable operation of the microcontroller even as the battery voltage drops.

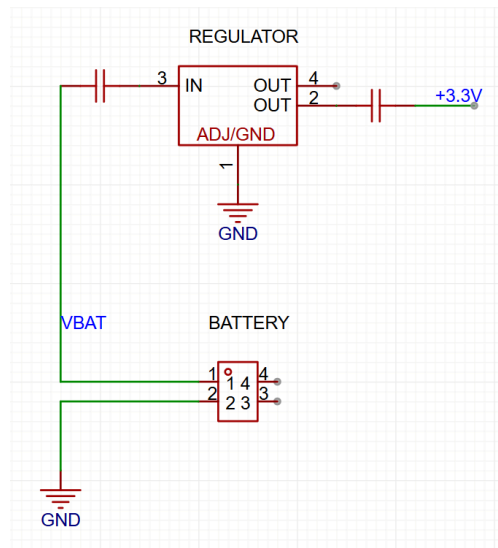


Figure 8. Battery with Regulator connection

5. PWM Motor Control

Motor speed control is achieved using PWM signals generated by Timer 1 (channels PA8 and PA9). To ensure smooth operation and avoid audible noise, a PWM frequency around 10 kHz is utilized. Directional control is handled through L293D input pins (IN_L1–IN_L2 and IN_R3–IN_R4). For example, during a left turn, the right motor operates at a higher PWM duty cycle (e.g., 80%), while the left motor either slows down or reverses its direction, depending on the desired turning radius.

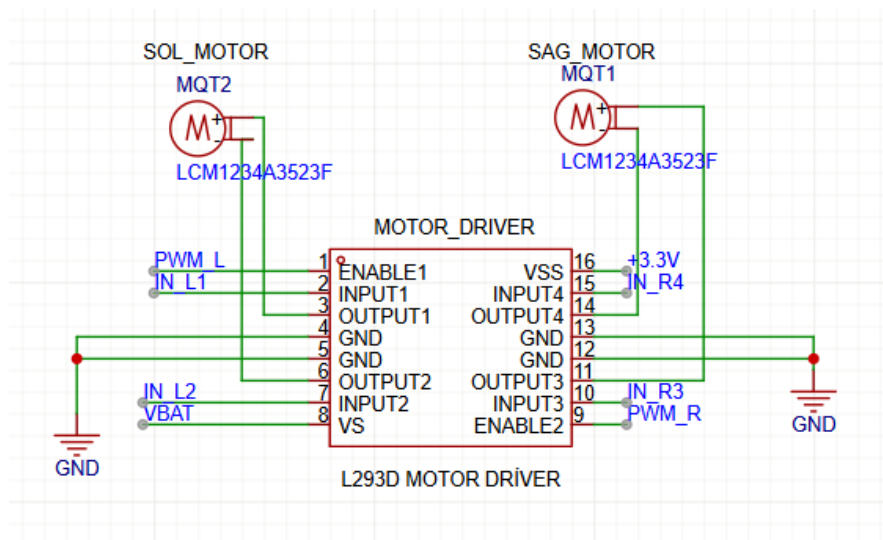


Figure 9. Motor Driver

6. Conclusion

This report has detailed the design and implementation of a line-following robot based on the STM32F103 microcontroller. Through the integration of optical sensors, PWM motor control, and ADC-based voltage monitoring, the robot demonstrates a complete embedded control system. Future improvements could include adaptive PID control for smoother line tracking and wireless telemetry for debugging and performance monitoring.

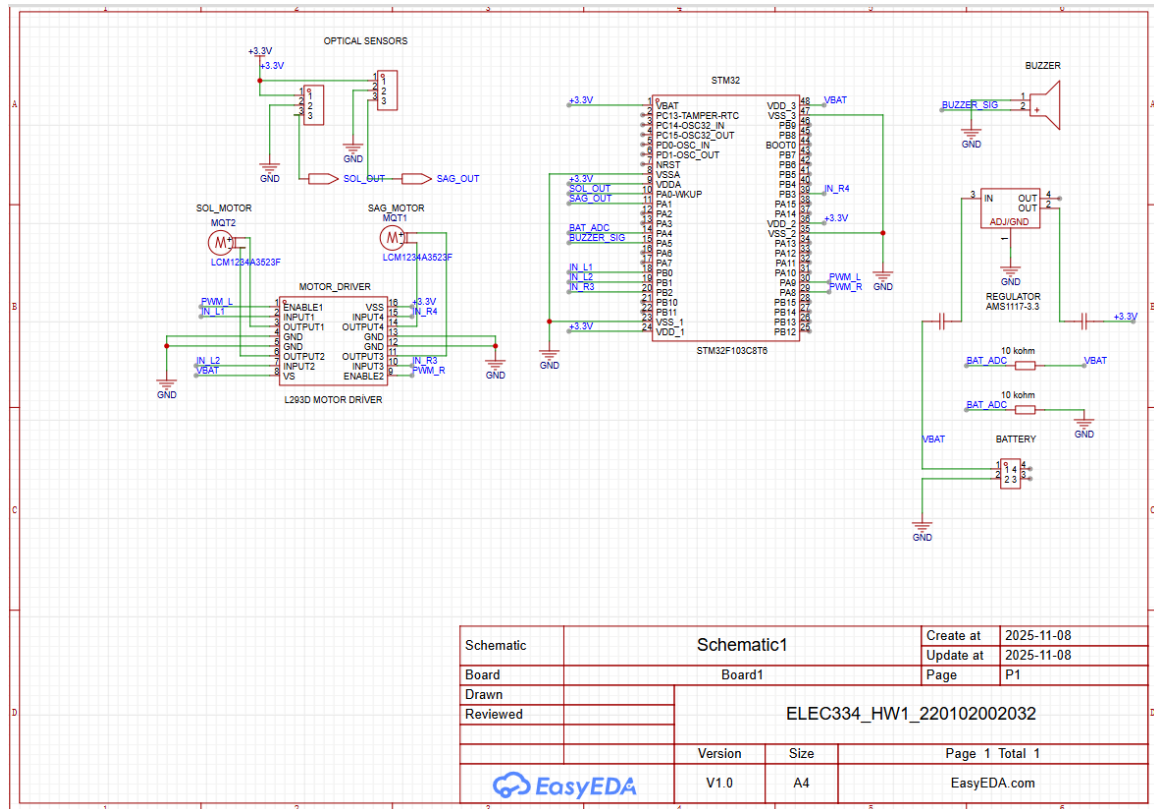


Figure 10.